Effective-sample-size calibration of a modality test for cluster homogeneity and the number of classes in spatially autocorrelated data

F. Parra [co-authors]

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Abstract

Selecting the number of clusters K is a basic step in unsupervised classification. The standard internal criteria (the gap statistic, the average silhouette, the Calinski–Harabasz and Davies–Bouldin indices) take no account of the dependence between observations, and so are miscalibrated on spatial data. There, neighbouring locations are autocorrelated, the effective sample size is far below the nominal one, and a smooth large-scale trend mimics discrete structure. We show that the consequence is systematic over-detection: on autocorrelated fields without discrete classes these criteria report spurious clusters in every one of 75 controlled realisations, and on real imagery they fragment single land-cover regions. We recast the choice of K as a recursive homogeneity test: at each step, whether a region is a single class or carries *multimodality in excess of spatial autocorrelation*. The proposed method, ESS-Dip, evaluates the Hartigan dip statistic along the locally optimal split direction. It calibrates the statistic against the effective sample size derived from a within-class autocorrelation range, which a local declustering estimator separates from the between-class scale; a low-order trend is removed adaptively. A recursive partitioning returns K, and the test carries an asymptotic false-split-control property. On a factorial simulation with known ground truth, ESS-Dip reaches unit specificity with competitive recovery; its recall-oriented variant ESS-Dip-R, calibrated against the model's Gaussian null, attains the best balanced score of sixteen criteria at the same specificity. On the Indian Pines, Salinas and Sentinel-2 scenes both identify single land-cover regions as homogeneous where the standard criteria do not. The method is conservative by design, and its negatives are reliable: when it reports a single class that verdict can be trusted, suiting ESS-Dip to use as a homogeneity guard or a stopping rule rather than as an exhaustive enumerator of weakly separated or strongly imbalanced classes, a trade-off we characterise. Code and data are released.

Keywords: number of clusters · cluster validity · spatial autocorrelation · effective sample size · declustering · unsupervised classification · remote sensing

1 Introduction

Unsupervised classification partitions a data set into groups without labelled examples, and in doing so it must answer a prior question: how many groups are there? The choice of the number of clusters K governs every downstream interpretation, and in the analysis of remotely sensed imagery, where K is the number of land-cover classes an analyst will map, it is made routinely and consequentially. A mature toolkit exists to make it: the gap statistic (Tibshirani et al., 2001), the average silhouette (Rousseeuw, 1987), the Calinski–Harabasz (Caliński and Harabasz, 1974) and Davies–Bouldin (Davies and Bouldin, 1979) indices each score a partition by its internal geometry and select the K that optimises the score. The stakes are concrete. An analyst mapping land cover from a satellite scene must fix K before any unsupervised classifier runs, and every downstream product (class areas, change statistics, sampling strata, training masks) inherits the choice. Set K too high and single crop fields fragment into spurious sub-classes that propagate into those products; set it too low and real classes merge. The same prior decision arises when delineating geochemical domains, zoning a mineral deposit, or partitioning a scene into regions for stratified field sampling: in each case the analyst must first ask how many classes the data actually support.

These criteria share a blind spot: none of them takes the dependence between observations into account. Their reference levels, the uniform null of the gap statistic and the implicit baseline of a silhouette, describe what the geometry would look like were the samples exchangeable draws. Spatial data are not exchangeable. Neighbouring pixels are autocorrelated (Tobler, 1970), so the effective number of independent observations is far below the nominal pixel count (Cressie, 1993), and a smooth illumination or topographic gradient spreads the feature cloud along a manifold that an independence null cannot distinguish from genuine groups. The consequence, which we quantify in Sections 4–5, is systematic *over-detection*(throughout, the partitions the criteria score are produced by k-means, the dominant choice in this setting; statements about a criterion's behaviour are therefore statements about the criterion–k-means pair, a dependence Section 4.2 makes explicit). On synthetic fields containing no discrete classes the gap statistic reports on average between seven and eight; on real imagery the silhouette, Calinski–Harabasz and Davies–Bouldin indices fragment a single, homogeneous crop field into several sub-classes in every instance we examined. A criterion that cannot recognise the absence of clusters is a precarious foundation for deciding their number.

Correcting this is not a matter of swapping in a better null. Because the within-cluster dispersion that these criteria minimise is a variance-based quantity, a reference matched to the data's spatial covariance structure, its variogram, reproduces the same dispersion and explains the clusters away. Distinguishing real classes from autocorrelation therefore requires information beyond the covariance, namely the *multimodality* of the feature distribution; and distinguishing real classes from a smooth trend requires separating a continuous drift from discrete structure, a separation the geostatistical decomposition into trend and residual makes natural. The idea of calibrating a clustering criterion against an autocorrelation-aware null is itself not new. Hennig and Lin (2015) do exactly this for discrete presence–absence data, but it has not been carried to the continuous-field setting of gridded imagery, nor has the confounding of trend with class been addressed in selecting K.

This paper supplies both. We recast the question *how many classes?* as a sequence of homogeneity decisions: at each node we test whether a region is a single class or instead carries *multimodality in excess of spatial autocorrelation*, and make that test valid on autocorrelated fields by calibrating it at the effective sample size. The construction rests on three geostatistical devices. The first is the effective sample size itself (Cressie, 1993; Dutilleul, 1993), which sets the calibration; it is evaluated at the indicator integral range appropriate to a functional of the distribution, not at the field integral range that governs the mean (Section 3.7). The second is a *local* estimate of the within-class autocorrelation range, in the manner of declustering (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998), which prevents the between-class mosaic from corrupting that calibration. The third is an adaptive trend–residual decomposition, which removes a smooth drift only when it is statistically resolvable. A recursive partitioning driven by the calibrated dip statistic (Hartigan and Hartigan, 1985) returns K. Because the test is tuned for specificity, its negatives are trustworthy: when it declares a region homogeneous that verdict can be relied on, which makes ESS-Dip as useful as a homogeneity guard or a stopping rule for region-growing as it is as a conservative estimator of K. Our contributions are:

- We document, on controlled ground truth, that the standard number-of-clusters criteria over-detect under spatial autocorrelation without exception, failing to recognise class-free scenes in every one of 75 realisations (Section 4).

- We propose *ESS-Dip*, an effective-sample-size-calibrated test of cluster homogeneity (whether a region holds discrete structure beyond its spatial autocorrelation), combining a within-class range from local range estimation with adaptive trend removal (Section 3). Applied recursively it returns a conservative K, and we state an asymptotic false-split-control property (Proposition 1) that the method confirms empirically. A recall-oriented variant, ESS-Dip-R, trades the distribution-free uniform null for the model's Gaussian null to recover more structure at the same specificity.

- We evaluate sixteen criteria on a factorial simulation with known K, including the most direct spatial adaptation of a standard index (the gap statistic with an autocorrelation-matched reference), including the calibrated-validity-index method of Hennig and Lin (2015) that ESS-Dip generalises; ESS-Dip reaches unit specificity, its recall variant ESS-Dip-R attains the best balanced score at the same specificity, and an ablation attributes the gain to the local range and the adaptive detrending (Section 4).

- We confirm the behaviour on three real scenes spanning hyperspectral and multispectral sensing (Indian Pines, Salinas, and a Sentinel-2 scene labelled by ESA WorldCover), where ESS-Dip identifies single land-cover regions as homogeneous where the standard criteria do not (Section 5).

The remainder of the paper is organised as follows. Section 2 reviews number-of-clusters criteria, the geostatistical notions of effective sample size and declustering, and modality-based clustering, and positions the contribution. Section 3 develops the method and its calibration. Sections 4 and 5 report the simulation study and the application to land-cover imagery. Section 6 discusses the precision–recall trade-off and limitations, and Section 7 concludes. Code and data are released to reproduce every result.

2 Background and related work

2.1 Selecting the number of clusters

Choosing the number of clusters K is a classical problem in unsupervised analysis, and a large family of *internal* criteria address it by scoring a partition against the geometry of the data alone. The gap statistic (Tibshirani et al., 2001) compares the within-cluster dispersion to its expectation under a reference distribution with no clustering, typically uniform over the data's bounding box; the average silhouette (Rousseeuw, 1987), the Calinski–Harabasz ratio (Caliński and Harabasz, 1974) and the Davies–Bouldin index (Davies and Bouldin, 1979) balance within- against between-cluster scatter. Comparative studies cover the many further variants (Arbelaitz et al., 2013). A second family selects K through a probabilistic model: model-based clustering fits a finite mixture and chooses the number of components by an information criterion such as the BIC (Gormley et al., 2023). None of these criteria was proposed with a distribution theory that requires independent observations. What they share is that the dependence between observations plays no role in their construction: the reference distribution of the gap statistic, the implicit baselines of the ratio indices, and the mixture likelihood treat the sample as exchangeable draws, and so ignore spatial dependence rather than model it. When observations are nearly independent the criteria are well calibrated; under strong dependence they are not. Spatial data are the archetypal failure, because nearby locations are correlated: "everything is related to everything else, but near things are more related than distant things" (Tobler, 1970).

2.2 Accounting for spatial dependence

That an independence reference over-detects structure in dependent data is known across several fields. In spatial epidemiology, Goovaerts and Jacquez (2004) show that a complete-spatial-randomness null over-identifies disease clusters and propose, in its place, a geostatistical *spatial neutral model* (realisations reproducing the histogram and variogram of the data by sequential Gaussian simulation), against which far fewer locations remain significant. In neuroimaging, Eklund et al. (2016) trace inflated false-positive rates in cluster-extent inference to a mis-specified parametric model of the spatial autocorrelation. The common remedy is a null that *preserves* the spatial covariance structure while removing the effect being tested, realised either by geostatistical simulation (Goovaerts and Jacquez, 2004) or by spatially constrained randomisation such as Moran spectral randomization (Wagner and Dray, 2015).

The same logic has been brought to bear on K itself. Hennig and Lin (2015) calibrate a validity index (the gap, silhouette or prediction strength) against its distribution under a problem-specific null that encodes non-clustering structure, including spatial autocorrelation, and use the calibrated index both to test for clustering and to estimate K; on a biogeographic example a plain Gaussian null favours eight clusters while the autocorrelation-aware null reduces this to two and renders the apparent clustering non-significant. Their framework is the conceptual basis of the present work. Its guiding principle, that all structure not attributable to clustering belongs in the null model, is exactly the principle the present method instantiates for spatially autocorrelated continuous fields; the two contributions differ in how the calibration is obtained and in the data setting they treat. First, Hennig and Lin (2015) calibrate by simulation, so every decision requires an ensemble of null datasets, whereas the calibration developed here is analytic, through the effective sample size, which is what makes the near-linear full-scene cost of Section 3.10 possible. Second, their construction is formulated for discrete presence–absence data with an algorithmic disjunction-parameter null, and addresses neither the *continuous-field* setting of gridded imagery nor the confounding of a smooth large-scale *trend* with discrete classes. The present contribution supplies exactly these: a Gaussian-random-field null on the lattice, a within-class range obtained by local range estimation, an adaptive trend–residual separation, and an effective-sample-size calibration of a modality statistic. Throughout the comparisons of Sections 4 and 5, the label Hennig–Lin accordingly denotes their published null-model construction applied to our setting, not this general framework, of which the proposed method is itself an instance.

The quantitative device that makes this possible is the *effective sample size*: under spatial autocorrelation the variance of a sample statistic behaves as if computed from $n_{\text{eff}} < n$ independent observations, a reduction governed by the integral of the autocorrelation (Cressie, 1993), a quantity formalised as the *integral range* in the theory of ergodic random functions (Lantuéjoul, 1991, 2002). The concept recurs across statistics whenever dependent data carry less information than their count suggests: Griffith (2005) develops effective geographic sample sizes for spatial sampling design, Berger et al. (2014) formalise the notion for general parametric inference, Vallejos and Osorio (2014) define and study it for spatial process models, and Acosta et al. (2021) extend its computation to large spatial datasets through block likelihoods. Dutilleul (1993) operationalised this idea to correct the degrees of freedom of a correlation test between two spatial processes; we use it to recalibrate, rather than to correct after the fact, the reference distribution of a clustering criterion. The autocorrelation scale it requires is the variogram range (Matheron, 1963; Cressie, 1993), and the safeguard against letting large homogeneous regions dominate its estimate is *declus-*

tering, a standard geostatistical correction for spatially clustered support (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998).

2.3 Modality as a clustering signal

Because k-means dispersion is itself a variance-based quantity, a null matched to the data's covariance structure can explain away the very clusters one seeks; a criterion that survives such a null must use information beyond the covariance. Multimodality is the natural choice. The dip test of Hartigan and Hartigan (1985) measures the departure of a univariate sample from unimodality, and dip-means (Kalogeratos and Likas, 2012) turns it into a divisive procedure that splits a cluster only when a projection of it is significantly multimodal, thereby estimating K. These methods assume independent observations; our contribution is to make the dip test valid under spatial autocorrelation by calibrating it at the effective sample size, and to pair it with the trend removal that prevents a smooth gradient from being read as two modes.

2.4 Distinction from spatially constrained clustering

A separate line of work *imposes* spatial contiguity on the partition: regionalization methods such as SKATER (Assunção et al., 2006), ClustGeo (Chavent et al., 2018) and the max-p-regions problem (Duque et al., 2012) constrain clusters to be spatially connected. These address a different question, how to partition space into contiguous regions, and still require the number of regions as an input or a tuning parameter. We do not constrain the partition; we correct the criterion that selects *how many* groups are present, and the two are complementary: the effective-sample-size calibration developed here could in principle inform the stopping rule of a regionalization just as it informs the recursion of Section 3.

3 Mathematical formulation

We formulate the selection of the number of clusters as a sequence of hypothesis tests for *multimodality in excess of spatial autocorrelation*. Section 3.1 fixes notation; Section 3.2 states the null against which a candidate split is judged; Sections 3.3–3.6 develop the test statistic, its effective-sample-size calibration, the within-class range estimator, and the trend–residual decomposition; Section 3.8 assembles these into a recursive partitioning algorithm; Section 3.9 states the resulting false-split control; and Section 3.10 discusses computation.

3.1 Setting and notation

Let $D \subset \mathbb{Z}^2$ be a finite regular lattice (the image domain) with $N = |D|$ pixels, and let

$$\boldsymbol{z} : D \to \mathbb{R}^p, \qquad \boldsymbol{z}(s) = \big(z_1(s), \ldots, z_p(s)\big)^\top,$$

be a multivariate field assigning to each site $s \in D$ a p-dimensional feature (for imagery, the spectral signature, whose components z_b, $b = 1, \ldots, p$, we call *bands*). The objects being clustered are the N feature vectors $\{\boldsymbol{z}(s)\}_{s \in D}$; the sites s carry the spatial dependence but are not themselves clustered. Unsupervised classification seeks a partition of D into K groups whose feature vectors are internally homogeneous. Our object of interest is the *number* of groups K, not the partition itself; the quality of the partitions induced along the way is nonetheless evaluated, against ground truth, in Section 5.4.

Table 1: Notation. Three decorrelation areas appear: the deployed method uses the ℓ^2 proxy, while a_{int} and a_I are the theoretical field and indicator integral ranges that Sections 3.7 and 3.9 relate to it.

$D,\ N$	image lattice and its pixel count
$\boldsymbol{z}(s) \in \mathbb{R}^p$	feature vector at site s; components z_b (bands)
$L,\ K,\ \hat{K}$	class labelling, true number of classes, its estimate
$\boldsymbol{c}_k,\ \boldsymbol{\mu}(s),\ \boldsymbol{\varepsilon}(s)$	class centres, smooth drift, stationary residual (1)
$\rho(\boldsymbol{h}),\ \boldsymbol{h}$	correlogram of $\boldsymbol{\varepsilon}$ at spatial lag $\boldsymbol{h} \in \mathbb{R}^2$
a	decorrelation area used by the calibration (5)
$\ell_j,\ \ell$	tile $1/e$-range and its tile median; default $a = \ell^2$ (7)
a_{int}	field integral range, (2) with a fitted variogram model
a_I	indicator integral range (Section 3.7)
$C,\ n_C,\ n_{\mathrm{eff}}$	candidate cluster, its pixel count, effective size n_C/a
$\boldsymbol{u},\ \hat{F}_C,\ \hat{D}_C$	split direction, empirical distribution function of the projection, its dip
$q_{1-\alpha}(m)$	$(1-\alpha)$ dip quantile for m independent draws
$\alpha,\ \tau,\ t,\ \nu,\ B$	level, detrend threshold, tile side, minimum n_{eff}, ensemble size

We model $\boldsymbol{z}$ as the superposition of a class component, a smooth drift and a stationary residual,

$$\boldsymbol{z}(s) = \boldsymbol{c}_{L(s)} + \boldsymbol{\mu}(s) + \boldsymbol{\varepsilon}(s), \tag{1}$$

where $L : D \to \{1, \ldots, K\}$ is a class labelling, $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_K \in \mathbb{R}^p$ are distinct class centres, $\boldsymbol{\mu}$ is a smooth large-scale drift (illumination, topography, atmospheric gradients), and $\boldsymbol{\varepsilon}$ is a zero-mean, weakly stationary residual carrying the short-range *within-class* autocorrelation. The three terms play distinct roles that the method must keep separate: the labelling L may be spatially arbitrary (no contiguity is assumed, although in imagery classes are typically contiguous); the drift $\boldsymbol{\mu}$ is smooth but carries no class information; and the autocorrelation is a property of the residual $\boldsymbol{\varepsilon}$ alone. The *true number of classes* is the number of distinct labels present, $K = |L(D)|$; a scene is *homogeneous* when $K = 1$, in which case (1) reduces to the classical geostatistical trend–residual decomposition. We denote by $\hat{K}$ the estimate returned by the recursive procedure of Section 3.8. The spatial dependence of $\boldsymbol{\varepsilon}$ is summarised by its correlogram $\rho(\boldsymbol{h})$, where $\boldsymbol{h} \in \mathbb{R}^2$ denotes the spatial lag vector, or equivalently its variogram, and condensed into a single scalar by the *decorrelation area*, the integral range (Lantuéjoul, 1991, 2002; Cressie, 1993),

$$a = \int_{\mathbb{R}^2} \rho(\boldsymbol{h})\, d\boldsymbol{h}, \tag{2}$$

the area occupied by one spatially independent observation. Estimating a from a fitted variogram is the subject of Section 3.5; for a subset $C \subseteq D$ we write $n_C = |C|$. Table 1 collects the notation; in particular, three related decorrelation areas appear in the paper, and the table states the role of each.

3.2 Clusters as excess multimodality

Internal validity criteria for K, namely the gap statistic (Tibshirani et al., 2001), the average silhouette, Calinski–Harabasz and Davies–Bouldin, compare a within-cluster dispersion against the value expected with *no* clustering, under an *independent* reference distribution. Two features of spatial data invalidate this comparison. First, autocorrelation reduces the number of independent observations far below N, so dispersion-based statistics are evaluated as if the sample were much larger than its information content (Cressie, 1993; Dutilleul, 1993). Second, a smooth drift $\boldsymbol{\mu}$ spreads the feature cloud along a low-dimensional manifold and is read as structure by an independence null. The combined effect is a systematic *over-detection* of clusters, which we document empirically in Section 4.

We therefore replace the dispersion comparison by a test of *modality*. Discrete classes manifest as a *multimodal* feature distribution; a single class (however autocorrelated, and however smoothly it varies in space) is *unimodal*. This motivates the reference hypothesis under which a candidate cluster is judged homogeneous.

Assumption 1 (Unimodal Gaussian-random-field null)**.** Under the null H_0 a cluster's residual field $\boldsymbol{\varepsilon}$ restricted to C is a weakly stationary Gaussian random field with decorrelation area a and *no* discrete component. In particular the marginal law of any fixed linear projection $\boldsymbol{u}^\top \boldsymbol{\varepsilon}$, $\boldsymbol{u} \in \mathbb{R}^p$, is univariate Gaussian, hence unimodal.

Assumption 1 is the continuous-field counterpart of the autocorrelation-aware null of Hennig and Lin (2015), who calibrated a validity index against a non-clustering reference for point data; here the reference is a Gaussian random field, the maximum-entropy unimodal model given the covariance structure, of the kind routinely generated by sequential Gaussian simulation (Goovaerts and Jacquez, 2004; Wagner and Dray, 2015).

3.3 The dip statistic along the split direction

Modality is quantified by Hartigan's dip (Hartigan and Hartigan, 1985). For a univariate sample with empirical distribution function F_m (size m), the dip is the supremum distance to the closest unimodal distribution,

$$\operatorname{dip}(F_m) = \inf_{G \in \mathcal{U}} \sup_{x \in \mathbb{R}} |F_m(x) - G(x)|, \tag{3}$$

$\mathcal{U}$ being the class of unimodal distribution functions. The dip is 0 for a perfectly unimodal sample and grows with the separation between modes.

Testing (3) on a one-dimensional projection requires a direction along which a genuine bimodality, if present, is exposed. Rather than the leading principal axis, which can hide modes spread across several dimensions, we test along the direction that a tentative binary split proposes. Given a cluster C, let k-means with $k = 2$ return centroids $\bar{\boldsymbol{z}}_0, \bar{\boldsymbol{z}}_1$, and set the unit split direction and projected sample

$$\boldsymbol{u} = \frac{\bar{\boldsymbol{z}}_1 - \bar{\boldsymbol{z}}_0}{\|\bar{\boldsymbol{z}}_1 - \bar{\boldsymbol{z}}_0\|}, \qquad y(s) = \boldsymbol{u}^\top \boldsymbol{z}(s), \quad s \in C. \tag{4}$$

The statistic is $\hat{D}_C = \operatorname{dip}(\hat{F}_C)$, where $\hat{F}_C$ is the empirical distribution function of $\{y(s)\}_{s \in C}$. Coupling the test to the split it must license makes the procedure sensitive to the bimodality k-means detects, while preserving its calibration under H_0: projecting a unimodal (Gaussian) cloud onto any fixed direction returns a unimodal sample, regardless of where k-means places the cut. The role of 2-means is thus confined to proposing the direction $\boldsymbol{u}$: the homogeneity decision itself is made by the modality test on the projection, which detects bimodality irrespective of unequal mode variances, and a node is actually partitioned only when the test licenses it. The direction remains a heuristic. k-means favours balanced, spherical groups, so when more than two classes coexist at a node, or modes differ strongly in variance, the proposed direction can fail to expose the multimodality or can cut through a class. The recursion re-tests each child, which recovers many such cases (the structured-world recovery of Section 4 spans K up to 5); a poor direction costs recall, never level (the fixed-direction check of Figure 1). Section 6 discusses mode-seeking directions as the natural refinement.

3.4 Effective-sample-size calibration

The null distribution of the dip depends on the sample size: for m *independent* draws from the least-favourable unimodal law (the uniform; Hartigan and Hartigan, 1985), $\mathrm{dip}(F_m) = O_p(m^{-1/2})$, and we denote by $q_{1-\alpha}(m)$ its $(1-\alpha)$ quantile. With autocorrelated data the nominal size n_C overstates the information content: the empirical distribution of n_C correlated values fluctuates about the population distribution at the rate of

$$n_{\text{eff}} = \frac{n_C}{a} \tag{5}$$

independent values, the effective sample size of a field with decorrelation area a (Cressie, 1993; Dutilleul, 1993). We therefore calibrate the observed dip against the null at the effective, not the nominal, size: the test rejects unimodality of C, and thereby licenses a split, when

$$\hat{D}_C \;>\; q_{1-\alpha}\big(\lfloor n_{\text{eff}} \rceil\big), \tag{6}$$

with $\lfloor\cdot\rceil$ the nearest integer. The quantile $q_{1-\alpha}(\cdot)$ has no closed form and is obtained by Monte Carlo: for each requested size m, first rounded to the nearest integer and clamped to $[12, 2000]$, the quantile is estimated from 300 simulated dips of uniform samples of size m, and the result is memoised, so each distinct size is simulated only once across the whole recursion and ensemble. The calibration therefore adds negligible computational cost. Autocorrelation thus raises the significance threshold (through a smaller n_{eff}) without discarding data: the statistic $\hat{D}_C$ is still computed from all n_C pixels, retaining full detection power.

The quantile $q_{1-\alpha}$ uses the *uniform* law, the least-favourable unimodal distribution (Hartigan and Hartigan, 1985): it bounds the level for any unimodal within-class law, but is conservative when that law is the Gaussian of Assumption 1. A recall-oriented variant, **ESS-Dip-R**, calibrates instead against the Gaussian null, drawing the reference dips from $\lfloor n_{\text{eff}} \rceil$ standard-normal samples; the two are the distribution-free and the model-matched ends of one calibration. The Gaussian reference is less conservative and recovers more structure while empirically preserving unit specificity (Sections 4.3 and 5.3); we report both and take the conservative uniform null as the default. Two caveats attend the Gaussian reference: it may reject a homogeneous but skewed or heavy-tailed within-class law, reading the asymmetry as a second mode, which is the price of its extra recall; and it is licensed by Assumption 1 rather than distribution-free. Conversely, the least-favourable property of the uniform null is an independent-sample result; what carries it to the effective-sample-size calibration under dependence is discussed in Section 3.9 (Remark 1).

3.5 Within-class decorrelation area by local range estimation

The calibration (5) requires the decorrelation area (2) of the *within-class* residual $\boldsymbol{\varepsilon}$, not of the between-class mosaic. Estimating it from the whole scene conflates the two: large contiguous classes inflate the apparent range, a grows, n_{eff} collapses, and the test loses all power, the field-scale rather than the within-class autocorrelation being measured. We avoid this conflation by estimating a *locally* over small tiles, the principle that also motivates declustering, where spatially clustered support is kept from dominating a statistic (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998).

Partition D into square tiles $\{T_j\}$ of side t pixels ($t \times t$ sites each). The premise is that most tiles fall *within* a single class, so that their autocorrelation structure reflects $\boldsymbol{\varepsilon}$ alone; we verify it empirically on the real scenes (62% of tiles are single-class on the contiguous Salinas and Sentinel-2 scenes, 27% on the fragmented Indian Pines fields, Section 5.2), and the median aggregation below supplies the robustness where it weakens. For each tile we take the first principal component of $\{\boldsymbol{z}(s)\}_{s\in T_j}$, compute its sample autocorrelation function as the inverse Fourier transform of the periodogram (the Wiener–Khinchin relation, at $O(t^2 \log t)$ cost per tile), and record the lag ℓ_j at which it first falls below $1/e$, with e the base of the natural logarithm: $\rho(\ell) = e^{-1}$ is the conventional definition of the correlation range, exact for the exponential model, averaged over the two axes. The within-class decorrelation area is the squared median range,

$$a = \ell^2, \qquad \ell = \mathrm{median}_j\ \ell_j, \tag{7}$$

the median being robust to the tiles that straddle a class boundary(it tolerates up to half the tiles being contaminated, and the single-class fractions quoted above make the straddling tiles a minority on all three real scenes), and the square turning a correlation length into the area of one decorrelation patch. We reuse the symbol a because this is the quantity the calibration (5) consumes; it is an operational proxy for a decorrelation area in the indicator-range regime of Section 3.7, not an estimate of the theoretical integral range (2) itself. This $1/e$-range proxy leaves the geometric constant relating range to area implicit. A more explicit route fits, by least squares, the best of an exponential, spherical and Gaussian variogram model (nugget c_0, partial sill c, range parameter r_j) to each tile and uses its closed-form integral range

$$a_j^{\mathrm{int}} \;=\; 2\pi\,\frac{c}{c_0+c}\int_0^\infty r\,\rho(r)\,dr \;=\; \kappa\,\frac{c}{c_0+c}\,r_j^2, \tag{8}$$

with $\kappa = 2\pi,\ \pi,\ 0.2\pi$ for the exponential, Gaussian and spherical models respectively; the radial integral contributes r_j^2, $r_j^2/2$ and $0.1\,r_j^2$ to the common angular factor 2π, and the nugget fraction $c_0/(c_0+c)$ enlarges the effective sample size. The fitted parameters (c_0, c, r_j) determine a_j^{int} in closed form; the scene value a_{int} aggregates the a_j^{int} over tiles by the same median as (7). This integral range is derived for the variance of the sample *mean*; applied to the dip, a functional of the empirical distribution, it over-corrects, and Section 4.5 finds it markedly more conservative than the ℓ^2 proxy, which we therefore retain as the default. An assumption-free alternative, the Gaussian-simulation reference, is taken up in Section 3.7.

3.6 Adaptive trend–residual separation

A smooth drift $\boldsymbol{\mu}$ is itself unimodal yet, left in place, enlarges ℓ and biases the test. We remove it adaptively. For each band b fit a low-order polynomial surface $\hat{\mu}_b(s) = \boldsymbol{\beta}_b^\top \boldsymbol{\phi}(s)$ by ordinary least squares, where $\boldsymbol{\phi}$ collects the monomials in the pixel coordinates up to degree two, and let

$$R^2_{\mathrm{poly}} = \frac{1}{p}\sum_{b=1}^{p}\left(1 - \frac{\sum_s \big(z_b(s) - \hat{\mu}_b(s)\big)^2}{\sum_s \big(z_b(s) - \bar{z}_b\big)^2}\right) \tag{9}$$

be the average variance explained by the drift. A genuine large-scale trend produces R^2_{poly} close to one, whereas a discrete mosaic, not well represented by a low-order polynomial, produces a small value. We therefore test the residual $\boldsymbol{\varepsilon} = \boldsymbol{z} - \hat{\boldsymbol{\mu}}$ in place of $\boldsymbol{z}$ whenever

$$R^2_{\mathrm{poly}} > \tau, \tag{10}$$

and test $\boldsymbol{z}$ unchanged otherwise. The rule is a decision with two error types, leaving a weak trend in place and needlessly detrending a trend-free scene; the threshold trades them off and cannot guarantee either outcome. Its value $\tau = 0.25$ is fixed once, from the trend world of the factorial design, and is not tuned per scene; the fixed-versus-adaptive detrending ablation of Section 4 quantifies the consequences of both error types. Detrending only when the drift is resolvable protects the power of the test on trend-free scenes, where detrending would needlessly strip signal. Equation (1) together with (10) is the universal-kriging decomposition restricted to the regime in which the drift is statistically resolvable.

3.7 A Gaussian-simulation reference

The effective-sample-size calibration (6) is an approximation: it maps an autocorrelated sample of size n_C to an independent sample of size n_{eff} and reads the dip quantile there. Two caveats attend it. First, the integral range (8) is derived for the variance of the sample *mean*, whereas the dip is a functional of the empirical *distribution*, governed by the level-crossing indicators rather than the field itself. For a Gaussian field the median-level indicator correlogram is the arcsin transform $\rho_I(\boldsymbol{h}) = \frac{2}{\pi} \arcsin \rho(\boldsymbol{h}) \leq \rho(\boldsymbol{h})$, a standard result of indicator geostatistics (Goovaerts and Jacquez, 2004), so the decorrelation area appropriate to the dip is the *indicator* integral range $a_I = \int_{\mathbb{R}^2} \rho_I \, d\boldsymbol{h}$, strictly below the field integral range ($a_{\text{int}} \approx 1.45\, a_I$ on our fields). This is the geostatistical reason the mean-based integral range over-corrects (Section 4.5). The $1/e$ proxy $a = \ell^2$ lies in the same indicator-range regime, within which the recovered $\hat{K}$ is insensitive to the precise area, so we retain it; Section 3.9 confirms the resulting level is controlled across the operating range of n_{eff}. The second caveat is that the calibration assumes isotropy. A reference that avoids both is the geostatistical *neutral model* (Goovaerts and Jacquez, 2004). By the spectral (fast-Fourier) method (Dietrich and Newsam, 1993; Lantuéjoul, 2002), simulate an ensemble of unconditional Gaussian realisations $\{\tilde{y}^{(b)}\}_{b=1}^{B}$ on the support of C, with covariance given by the fitted variogram of the projection y; unimodality is rejected when

$$\hat{D}_C \;>\; q_{1-\alpha}\big(\{\operatorname{dip}(\tilde{y}^{(b)})\}_{b=1}^{B}\big). \tag{11}$$

For a *single* homogeneity decision on a full rectangular domain, testing $K = 1$ against $K \geq 2$, this Monte Carlo test makes no independence approximation and accommodates anisotropic or nested structure, and we show in Section 4.5 that it decides correctly where the effective-sample-size shortcut requires its calibration constant. Two limitations keep it from supplanting (6) as the estimator, however. It is exact only up to the *plug-in* variogram, fitted from the same data; and when iterated over the recursion its per-node variogram must be fitted on the non-rectangular support of a candidate cluster, and the repeated tests incur multiple comparisons. We implement the resulting recursive estimator (Section 4.5): a direct version over-splits, a multiple-testing correction makes it competitive with (6) on controlled fields, but its plug-in variogram on irregular supports keeps it behind on real imagery, and at the cost of B simulations per node against the cached, near-linear cost of (6). We therefore retain (6) as the method and use (11) as an assumption-light check on the leading split.

3.8 Recursive partitioning

The components above are combined into a top-down recursion (Algorithm 1). Beginning with $C = D$, a cluster is split into its 2-means children whenever the calibrated dip test (6) is significant and both children retain at least one decorrelation patch ($n \geq a$, so that $n_{\text{eff}} \geq 1$); otherwise it becomes a leaf. The estimate $\hat{K}$ is the number of leaves. To absorb the variability of the k-means initialisation the recursion is repeated B times with independent seeds and the median of $\{\hat{K}^{(b)}\}_{b=1}^{B}$ is reported(the median rather than the mode: the two coincide when the small integer-valued ensemble concentrates, and the median resolves split ensembles without a tie-breaking rule).

Algorithm 1 ESS-calibrated modality test for K (single ensemble member)

Require: field $\boldsymbol{z}$ on D; level α; degree-2 drift threshold τ; tile side t; minimum effective size ν

if $R^2_{\text{poly}}(\boldsymbol{z}) > \tau$ **then** ▷ Eq. (9)–(10)
    $\boldsymbol{z} \leftarrow \boldsymbol{z} - \hat{\boldsymbol{\mu}}$ ▷ trend removal
end if
$a \leftarrow (\text{median}_j\, \ell_j)^2$ ▷ local $1/e$-range area, Eq. (7)
stack $\leftarrow \{D\}$; leaves $\leftarrow 0$
while stack $\neq \emptyset$ **do**
    pop C from stack; $n_{\text{eff}} \leftarrow n_C/a$
    if $n_{\text{eff}} < \nu$ **then**
        leaves $\leftarrow$ leaves $+1$; **continue** ▷ insufficient independent information
    end if
    $(\bar{\boldsymbol{z}}_0, \bar{\boldsymbol{z}}_1) \leftarrow$ 2-MEANS(C); $\boldsymbol{u} \leftarrow (\bar{\boldsymbol{z}}_1 - \bar{\boldsymbol{z}}_0)/\|\bar{\boldsymbol{z}}_1 - \bar{\boldsymbol{z}}_0\|$
    $\hat{D}_C \leftarrow \text{dip}\big(\{\boldsymbol{u}^\top \boldsymbol{z}(s)\}_{s\in C}\big)$ ▷ Eq. (3)–(4)
    if $\hat{D}_C > q_{1-\alpha}(\lfloor n_{\text{eff}} \rceil)$ **and** both children have $n \geq a$ **then**
        push both 2-means children onto stack ▷ Eq. (6)
    else
        leaves $\leftarrow$ leaves $+1$
    end if
end while
return $\hat{K} =$ leaves

3.9 False-split control

The calibration (6) is designed so that, in the absence of discrete structure, the recursion does not split. We give conditions under which the per-node test controls its level asymptotically. The result is deliberately stylised: it concerns the test itself, with the projection direction held fixed, the detrending step of Section 3.6 switched off, and the decorrelation area equal to the indicator integral range a_I. The distance between this stylised setting and the deployed procedure is made explicit in Remark 1 and quantified empirically below. Write (C1) for the assumption that on C the residual $\boldsymbol{\varepsilon}$ is a stationary Gaussian random field whose correlogram ρ is nonnegative and absolutely summable, $\sum_{\boldsymbol{h}} \rho(\boldsymbol{h}) < \infty$, and (C2) for the projection direction $\boldsymbol{u}$ being fixed. Write (C3) for the hypothesis that the projected empirical process obeys a functional central limit theorem at the indicator integral range,

$$\sqrt{n_C/a_I}\,\big(\hat{F}_C - \Phi\big) \;\rightsquigarrow\; \mathbb{B}\circ\Phi \qquad \text{in } \ell^\infty(\mathbb{R}), \tag{12}$$

where $\rightsquigarrow$ denotes weak convergence (van der Vaart and Wellner, 1996, Ch. 1), Φ is the projected Gaussian law, $\mathbb{B}$ is the standard Brownian bridge, and $a_I = \int_{\mathbb{R}^2} \frac{2}{\pi} \arcsin \rho(\boldsymbol{h})\, d\boldsymbol{h}$ is the indicator integral range of Section 3.7. Hypothesis (C3) is the exact formal statement of the effective-sample-size heuristic for a distributional statistic: it asserts that the empirical process of the n_C autocorrelated observations behaves, to first order, like that of n_C/a_I independent draws from Φ.

Proposition 1 (Asymptotic false-split control)**.** *Assume* (C1)–(C3)*, let C increase to $\mathbb{Z}^2$ along a sequence of rectangles (increasing-domain asymptotics: $n_C \to \infty$ with the correlogram ρ held fixed), and calibrate the test* (6) *at $n_{\text{eff}} = n_C/a_I \to \infty$. Then*

$$\limsup_{n_C\to\infty}\; \mathbb{P}\Big(\hat{D}_C > q_{1-\alpha}\big(\lfloor n_{\text{eff}} \rceil\big)\Big) \;\leq\; \alpha, \tag{13}$$

and the limit is in fact zero. Consequently, if every node visited by the recursion satisfies (C1) *(the scene is a single class), then* $\mathbb{P}(\hat{K} = 1) \to 1$.

Proof. The argument reduces the dependent problem to the independent one; the spatial dependence enters only through (C3), and the properties of the dip that are used involve no assumption on the dependence structure of the data. First, since Φ is unimodal, the deterministic inequality $\hat{D}_C = \mathrm{dip}(\hat{F}_C) \leq \sup_x |\hat{F}_C(x) - \Phi(x)|$ (Hartigan and Hartigan, 1985) together with (12) shows that $\sqrt{n_{\text{eff}}}\,\hat{D}_C$ is tight: the dip vanishes at least at the $n_{\text{eff}}^{-1/2}$ rate. Second, at a strictly unimodal law the rate is strictly faster. Fluctuations of size ϵ about a distribution function with strictly curved branches can be absorbed into the unimodal class to within $o(\epsilon)$, an absorption that uses only the uniform continuity of the limiting process: for independent samples from a strictly unimodal density, smooth and with negative curvature at its mode, Cheng and Hall (1998) establish $D_m = O_p(m^{-3/5}) = o_p(m^{-1/2})$, the Gaussian satisfying their conditions, and under (12) the same absorption gives $\sqrt{n_{\text{eff}}}\,\hat{D}_C \to 0$ in probability. Third, the calibration quantile inherits $q_{1-\alpha}(m) = c_\alpha m^{-1/2}(1 + o(1))$, with $c_\alpha > 0$, from the uniform law, whose rescaled dip has a nondegenerate limit and which is asymptotically least favourable among unimodal laws (Hartigan and Hartigan, 1985). Hence $\mathbb{P}\big(\hat{D}_C > q_{1-\alpha}(\lfloor n_{\text{eff}} \rceil)\big) \to 0$, which gives (13); for a general unimodal projected law the least-favourable property caps the limit at α, attained only in the flat-topped boundary case, where linear stretches of the distribution function leave no curvature to absorb the fluctuations. Under the single-class hypothesis every node visited by the recursion satisfies (C1); the recursion visits finitely many nodes, so a union bound over their (vanishing) rejection probabilities gives $\mathbb{P}(\hat{K} = 1) \to 1$. □

Remark 1. Four observations delimit what the proposition does and does not establish. (i) *Condition (C3) carries the analytic weight.* It is the empirical-process counterpart, at the indicator integral range, of the effective-sample-size heuristic. It is precisely the step that a formal justification of "plugging an effective sample size into the independent reference distribution" requires for a distributional statistic such as the dip; for the dip the appropriate decorrelation scale is the indicator range a_I, not the field integral range that governs the sample mean (Section 3.7). Functional central limit theorems of this form are established for empirical processes of strongly mixing and of associated sequences and fields, and (C1) places ε in the associated class through Pitt (1982); Bulinski and Shashkin (2007); a complete verification for the lattice empirical process under our exact conditions is beyond the present scope, so we state (C3) as a hypothesis rather than derive it. (ii) *Why the level is conservative rather than exact.* The critical value is calibrated on the least-favourable uniform null, whereas under (C1) the projected data are Gaussian, whose rescaled dip vanishes asymptotically; the rejection probability therefore tends to zero, not to α. This is consistent with the zero empirical false-split rate of Figure 1a, and exact level α would require calibrating on the model's own null, which is what the recall variant ESS-Dip-R does, at the price of leaving the distribution-free guarantee. (iii) *The deployed default is covered empirically, not by the proposition.* The proposition takes the decorrelation area to be a_I; the deployed default estimates it by the ℓ^2 proxy of Section 3.5, which in our experiments lies below a_I ($\ell^2 \approx 0.3$–$0.56\,a_I$) and therefore *overstates* the effective sample size, pushing the test towards anti-conservatism. It is the uniform-null slack of point (ii) that absorbs this bias, and the net level of the deployed pipeline is what Figure 1 and the single-class windows of Section 5.3 verify empirically. The same accounting explains why ESS-Dip-R, which removes the uniform slack, is not licensed by the proposition and is validated on the model's null only. (iv) *Two further idealisations.* The detrending step is switched off, and under (C1) this costs no generality asymptotically. There is no deterministic trend under the null, and the adaptive rule of Section 3.6 is asymptotically inactive: the quadratic design spans a fixed number of low-frequency directions, so the variance it captures is bounded by a constant multiple of $\sigma^2 a$ (each eigenvalue of the residual's covariance matrix is at most $\sum_{\boldsymbol{h}} |C(\boldsymbol{h})| = \sigma^2 a$, finite precisely because (C1) makes the correlogram summable), while the total sum of squares grows in proportion to $n_C \sigma^2$. Hence $R^2_{\text{poly}} = O_p(1/n_{\text{eff}})$ and $\mathbb{P}(R^2_{\text{poly}} > \tau) \to 0$: the detrending trigger switches off at exactly the rate at which the calibration becomes valid. Under long-range dependence, where summability fails, this argument, like the calibration itself, would break. And the split direction is data-dependent in practice, (C2) being the fixed-direction idealisation under which Figure 1 shows no level inflation when the 2-means direction is used instead. Non-Gaussian within-class laws fall outside (C1) altogether, where the distribution-free uniform null of ESS-Dip is the safer reference.

The Gaussian-simulation reference (11) controls the level exactly for a single homogeneity test and so provides an assumption-light check on the leading split, but, as Section 4.5 shows, it does not yield a better recursive estimator. The effective-sample-size test (6) is thus an approximation whose adequacy is assessed empirically: Section 4 finds that, over the 75 null and trend realisations with no discrete classes, the procedure returns $\hat{K} = 1$ in every case, whereas the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices report spurious clusters in all of them.

The calibration can be probed directly, in isolation from the recursion. The design is a single homogeneity decision per scene (root node only): one-class Gaussian random fields with kernel bandwidths $\sigma \in \{2, 4, 6, 8\}$ on square domains of side 32 to 80 pixels, 160 realisations per cell, spanning effective sample sizes from 35 to over 300 (the regime the recursion visits). The power panel uses two-class fields of the same sizes at separation $\delta = 3$, the middle of the factorial design's range; the power curve is therefore specific to that separation and shifts with it. On these single-class fields the per-node test of Section 3.4 never produces a false split: its empirical false-split rate is 0.00 at every n_{eff} (Figure 1a), at or below the nominal α, and unchanged whether the projection direction is fixed or chosen by 2-means, so the data-dependent direction does not inflate the level. Calibrating instead at the nominal size n_C breaks this control (open symbols). The conservatism is not vacuous: on two-class fields the power rises with n_{eff} to one (Figure 1b). The effective size borrowed from sample-mean theory therefore calibrates the dip across the operating range, holding the level conservatively, even though its exact constant for the dip functional is left undetermined. The zero, rather than merely at-most-α, false-split rate is the deliberate conservatism of the least-favourable calibration that Remark 1(ii) explains, and it is what the recall variant ESS-Dip-R trades away for power (Section 3.4).

3.10 Computational considerations

The range estimator (7) costs one fast Fourier transform per tile, $O(N \log N)$ in total. Each visited node runs one 2-means fit, of cost $O(n_C p)$ with mini-batch updates, and one dip evaluation, $O(n_C \log n_C)$ after sorting the projection; the Monte Carlo quantiles $q_{1-\alpha}(\cdot)$ are computed once per effective-size bucket and reused. As the recursion visits each pixel in $O(\hat{K})$ nodes, one ensemble member is near-linear in N, and the B members are independent and trivially parallel. The procedure therefore scales to full scenes; Section 4.6 reports timings confirming a near-linear empirical exponent and a memory footprint a small multiple of the data, together with a constant-factor advantage over the gap statistic, whose reference replicates multiply its cost.

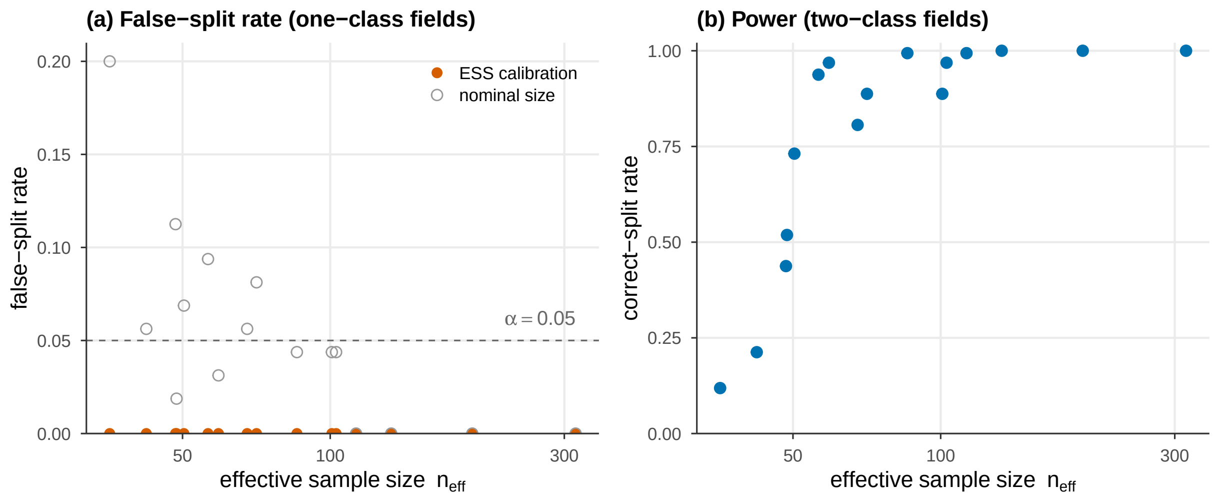

Figure 1: Direct validation of the effective-sample-size calibration on a single homogeneity decision, 160 realisations per cell. (a) On one-class Gaussian fields the false-split rate of the calibrated test stays at 0.00, at or below the nominal $\alpha = 0.05$, across effective sample sizes from 35 to over 300; calibrating at the nominal size instead exceeds α (open symbols). (b) On two-class fields the power rises with n_{eff} to one, so the level control is not the trivial conservatism of a test that never rejects.

4 Simulation study

We assess the procedure on synthetic scenes with known ground truth, where the number of classes, the strength of spatial autocorrelation, the presence of a trend, and the class separability are all controlled. The design isolates the two failure modes of Section 3.2, autocorrelation and large-scale drift, and lets us verify the false-split control of Proposition 1 directly, by counting how often each method recovers a single class when the scene contains exactly one.

4.1 Generative model

All scenes live on a 96×96 lattice with $p = 6$ bands. The elementary building block is a unit-variance Gaussian random field g_σ, obtained by convolving Gaussian white noise with an isotropic Gaussian kernel of bandwidth σ (which sets the autocorrelation range) and standardising. Four world types instantiate the decomposition (1):

Null ($K = 1$): each band an independent g_σ, $\sigma \in \{3, 6, 9\}$. Pure autocorrelation, no discrete structure.

Trend ($K = 1$): each band $\theta\, t(s) + 0.4\, g_3(s)$, with t a standardised planar–bilinear polynomial surface and trend amplitude $\theta \in \{1.5, 3.0\}$. Smooth large-scale variation, no classes.

Structured ($K \in \{2, 3, 4, 5\}$): a contiguous label map is drawn as the pixelwise $\arg\max$ of K independent fields g_{σ_f}, $\sigma_f \in \{6, 9\}$; class k is assigned a mean spectrum $\boldsymbol{\mu}_k \sim \mathcal{N}(\mathbf{0}, \delta^2 I_p)$ with separation $\delta \in \{2, 3, 4\}$, and a per-band autocorrelated noise g_2 is added. Discrete classes embedded in autocorrelated noise.

Mixed ($K \in \{3, 4\}$): a structured scene ($\delta = 3$, $\sigma_f = 6$) plus a global polynomial trend of amplitude $\theta \in \{1.5, 3.0\}$. The realistic case: classes *and* drift together.

The factorial of these settings yields 33 cells (3 null, 2 trend, 24 structured, 4 mixed); each cell is realised 15 times with independent seeds, for 495 scenes in total.

4.2 Methods and metrics

We compare sixteen criteria; the exact decision rule of each is given here or in Section 3, and every implementation, with its seeds, is in the released code. Four are standard internal indices evaluated over $K = 1, \ldots, 8$ on MiniBatch k-means partitions: the gap statistic (Tibshirani et al., 2001) with eight uniform reference datasets over the bounding box and the one-standard-error rule, the average silhouette, Calinski–Harabasz, and Davies–Bouldin (each maximised, or minimised for Davies–Bouldin, over K). These four are the original formulations, with no null model added by us; the null-calibrated criteria in this benchmark are the spatial gap next, the Hennig–Lin variants, and the proposed family, and in each case the null is an addition of the respective later work, not of the original index papers. A fifth is the most direct spatial adaptation of an existing index, the *gap with an autocorrelated reference*: identical to the gap statistic except that the eight reference datasets are drawn not uniformly over the bounding box but as per-band Gaussian random fields whose $1/e$ range matches the tile-median estimate of Section 3.5, scaled to each band's standard deviation, with the same one-standard-error rule. It isolates how far correcting only the reference's autocorrelation goes. (The latter three cannot return $K = 1$ by construction, an asymmetry we return to below.) Two more are the calibrated-validity-index method of Hennig and Lin (2015), the procedure ESS-Dip extends: the average silhouette width calibrated by parametric bootstrap against a null model, run both with an *i.i.d.* Gaussian null (the autocorrelation-blind application) and with a *spatial* null (phase-randomized surrogates preserving each band's variogram, their autocorrelation-aware variant adapted to the gridded setting). The calibrated index is $V(K) = \big(\mathrm{ASW}(K) - \overline{\mathrm{ASW}}_{\mathrm{null}}(K)\big)/\mathrm{sd}_{\mathrm{null}}(K)$, computed from $B = 20$ null datasets (scenes subsampled to 1200 pixels for tractability); $\hat{K}$ is the maximiser of V when $\max_K V(K) > 1.96$ and $\hat{K} = 1$ otherwise. An eighth baseline, plain *dip-means*, applies the same modality test at the nominal sample size, without the effective-sample-size calibration, the local range, or the trend removal, isolating the contribution of that spatial machinery. The ninth is the proposed method of Section 3, denoted **ESS-Dip**: Algorithm 1 with the local within-class range (7), adaptive trend removal (10), and a $B = 5$ ensemble; we fix $\alpha = 0.05$, $\tau = 0.25$, tile side $t = 24$, and $\nu = 12$ throughout, with no per-scene tuning. The tenth, **ESS-Dip-R**, is its recall-oriented variant: it calibrates the dip against the Gaussian null of Assumption 1 rather than the uniform least-favourable null, which is less conservative and recovers more structure at the same specificity (Section 4.3). The remaining four are ablations of ESS-Dip that disable one component each, to attribute the contribution of every ingredient: *global range* (the ℓ^2 range estimated over the whole image rather than as a tile median); *no trend removal* ($\tau = \infty$); *forced trend removal* ($\tau = 0$); and *declustering*, the precursor that subsamples the scene to one pixel per decorrelation patch and applies the dip test to the thinned, approximately independent sample. The fifteenth, *SGS-Dip*, replaces the per-node effective-sample-size calibration by the recursive Gaussian-simulation reference (11) of Section 3.7 ($B = 40$ simulations per node); the sixteenth adds a Bonferroni multiple-testing correction to it.

Three metrics summarise performance. *Specificity* is the fraction of null and trend scenes (the 75 scenes with $K = 1$) for which $\hat{K} = 1$; it is the empirical analogue of Proposition 1. *Exact-K accuracy* is the fraction of scenes with $\hat{K} = K_{\mathrm{true}}$, reported separately for the structured and mixed worlds, with the mean absolute error $\mathrm{MAE} = \mathbb{E}\,|\hat{K} - K_{\mathrm{true}}|$. The *balanced score* is the mean of specificity, structured accuracy and mixed accuracy, a single figure that rewards methods strong across all three regimes rather than in one.

Table 2: Performance over the 495 synthetic scenes. Specificity is the $\hat{K}$=1 rate on the 75 class-free (null+trend) scenes; accuracy is the exact-K rate. Best in each column in bold.

Method	Specificity ($\hat{K}$=1)	Structured Acc.	Structured MAE	Mixed Acc.	Balanced score
Standard internal indices (independence null)					
Gap statistic	0.00	**0.97**	**0.03**	0.03	0.336
Silhouette	0.00	0.76	0.32	0.30	0.355
Calinski–Harabasz	0.00	0.85	0.18	0.40	0.418
Davies–Bouldin	0.00	0.70	0.41	0.20	0.299
Calibrated validity index (Hennig and Lin, 2015)					
i.i.d. null	0.00	0.67	0.42	0.17	0.280
spatial null	0.80	0.72	0.38	0.17	0.561
Spatial adaptation of an existing index					
Gap (autocorrelated ref.)	0.21	**0.97**	0.04	0.10	0.427
Modality test without spatial calibration					
Dip-means (nominal)	0.52	0.92	0.09	0.02	0.486
Proposed					
ESS-Dip	**1.00**	0.81	0.35	0.50	0.771
ESS-Dip-R (recall)	**1.00**	0.89	0.19	**0.72**	**0.868**
Ablations of ESS-Dip					
global range	1.00	0.63	0.87	0.28	0.638
no trend removal	1.00	0.63	0.89	0.13	0.587
forced trend removal	1.00	0.19	1.95	0.30	0.498
declustering	1.00	0.63	0.83	0.08	0.572
Exact reference as a recursive estimator					
SGS-Dip	0.85	0.49	0.91	0.27	0.535
+ multiple-testing correction	0.99	0.79	0.26	0.35	0.710

4.3 Results

Table 2 reports the three metrics; Figure 2 displays the specificity–accuracy trade-off and the decay of accuracy with the true number of classes.

Standard indices over-detect without exception. On the 75 class-free scenes every standard index reports spurious structure: the $\hat{K}$=1 rate is 0.00 for all four. The failure is severe, not marginal: the gap statistic returns on average $\hat{K} = 7.7$ classes on fields that contain none, and the silhouette, Calinski–Harabasz and Davies–Bouldin indices average 2.9, 3.6 and 4.1. This is the over-detection anticipated in Section 3.2, now quantified: under spatial autocorrelation an independence null cannot recognise the absence of clusters. Nor is the failure repaired by the simplest spatial correction: the gap with an autocorrelation-matched reference lifts specificity only to 0.21, because matching the reference's autocorrelation addresses one of the two failure modes but not the other, the smooth drift. On trend scenes its reference fields carry no drift and the criterion reads the gradient as structure ($\hat{K} \approx 7$ on average), and in the mixed regime its accuracy collapses to 0.10;

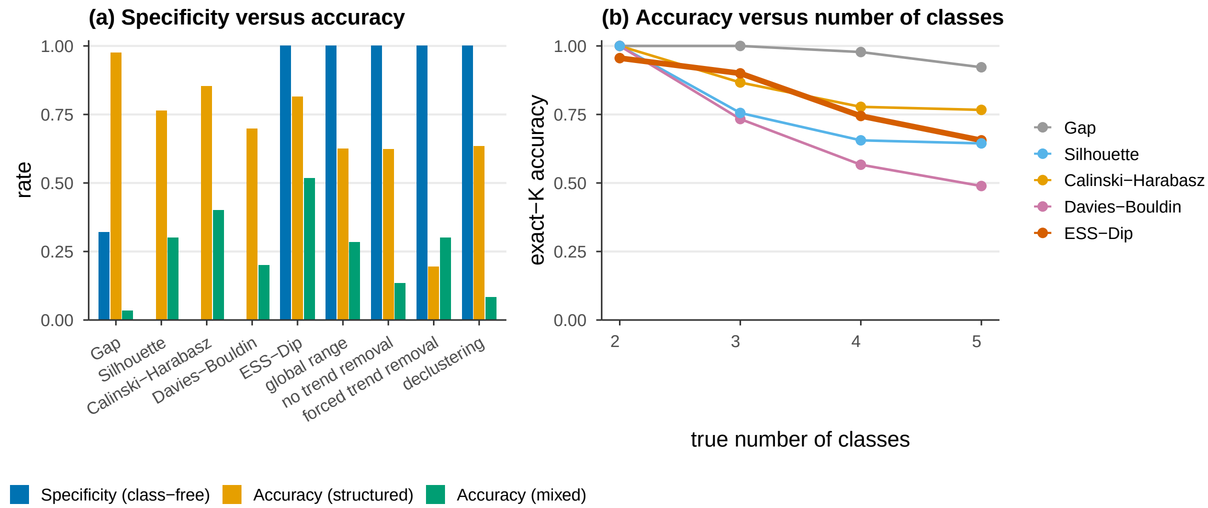

Figure 2: Left: specificity (class-free scenes) against exact-K accuracy on structured and mixed scenes, by method. Only the proposed family attains unit specificity; among them ESS-Dip recovers the most structure. Right: exact-K accuracy versus the true number of classes on structured scenes. For legibility the panels show the four standard indices and the ESS-Dip family; the calibrated-validity-index and Gaussian-simulation methods of Table 2 are omitted.

its balanced score, 0.427, sits far below the proposed family's. Fixing the reference is not enough; the dispersion statistic itself, and the trend, must both be addressed.

The proposed family controls the false-split rate. Every member of the proposed family, ESS-Dip and all four ablations, attains specificity 1.00: in 75 out of 75 class-free scenes it returns a single class. This is the empirical content of Proposition 1, and it holds whether the class-free scene is stationary (null) or carries a smooth drift (trend), confirming that the effective-sample-size calibration and the adaptive detrending together neutralise both failure modes.

Specificity does not cost competitiveness. On the structured scenes the gap statistic is most accurate (0.97): with well-separated classes and an aligned null it is hard to beat, and we do not. ESS-Dip reaches 0.81, ahead of the silhouette (0.76), Davies–Bouldin (0.70) and close to Calinski–Harabasz (0.85), while being the *only* method with non-zero specificity. On the realistic mixed scenes ESS-Dip reaches 0.50, ahead of every standard index (0.40 for the best), because the adaptive detrending removes the drift that collapses them. The recall variant ESS-Dip-R does better still: calibrating against the Gaussian null of Assumption 1 rather than the conservative uniform one, it raises structured accuracy to 0.89 and mixed accuracy to 0.72 at the same unit specificity, for a balanced score of 0.868, the best of any method. ESS-Dip itself scores 0.771, already far above the next non-proposed method (0.638) and more than twice the best standard index (0.418). These synthetic fields are Gaussian, so the model-matched ESS-Dip-R is favoured here; its specificity and recall gains carry to the real scenes of Section 5. Standard indices that look strong on pure structured scenes collapse once specificity is taken into account.

Comparison with the method ESS-Dip extends. The calibrated-validity-index framework of Hennig and Lin (2015) is instructive precisely because the present method generalises it. Applied with an independence (i.i.d.) null it inherits the over-detection of the standard indices (specificity 0.00, balanced score 0.280), confirming that the framework alone, without a spatial null, does not address the problem. With its autocorrelation-aware (spatial) null it recovers much of the lost specificity (0.80) and a comparable structured accuracy (0.72), a clear vindication of its central idea. Two differences remain, and they are the contribution of this paper. First, its specificity, at 0.80, still admits one class-free scene in five as clustered, where the within-class calibration of ESS-Dip admits none. Second, and decisively, on the mixed regime its accuracy is only 0.17, no better than the i.i.d. variant and far below ESS-Dip's 0.50, because a phase-randomized null preserves the trend along with the autocorrelation and so cannot separate a smooth drift from discrete classes. The explicit trend–residual decomposition of Section 3.6, absent from the calibrated-index framework, is what closes this gap. This comparison is insensitive to the significance threshold: the value used, $V > 1.96$ (the one-sided 2.5% level for the calibrated index under approximate normality), and the alternatives $V > 1.64$ and $V > 2.33$ all leave the conclusion intact: the i.i.d. null fails specificity (0.00) at every threshold, and the spatial null controls specificity (0.77–0.83) but not the mixed regime (≤ 0.08) at every threshold. Figure 3 makes the mixed-scene behaviour concrete on one realisation: the gap statistic and the Hennig–Lin null fragment the scene into six and eight clusters, whereas ESS-Dip recovers the three true classes.

4.4 Ablation

The ablations isolate where ESS-Dip's accuracy comes from (Table 2, lower block). The single largest contribution is the *local within-class range* of Section 3.5: replacing it by a scene-wide range drops structured accuracy from 0.81 to 0.63 and mixed accuracy from 0.50 to 0.28. This confirms the diagnosis of Section 3.5 (a global range conflates the between-class mosaic with the within-class scale, inflating ℓ, collapsing n_{eff}, and starving the test of power) while leaving specificity untouched. *Adaptive* trend removal matters in both directions: forcing detrending on every scene ($\tau = 0$) destroys structured accuracy (0.19), because the polynomial absorbs genuine between-class signal when no trend is present; disabling it ($\tau = \infty$) halves mixed accuracy (0.13), because an unremoved drift inflates the range. Removing the drift *only when present* attains both (0.81 structured, 0.50 mixed). The declustering precursor, which discards all but one pixel per patch, matches the others on specificity but trails on accuracy (0.08 mixed), confirming the value of calibrating on the full pixel set rather than thinning it. At the opposite extreme, plain dip-means (Table 2) removes the spatial apparatus entirely and applies the dip at the nominal size: the modality signal alone recovers structure well (0.92, second only to the gap statistic), but without the effective-sample-size calibration its specificity falls to 0.52, and without trend removal its mixed accuracy collapses to 0.02. The contribution of ESS-Dip is therefore not the modality test, which is already strong, but the spatial calibration and trend separation that turn it from a high-recall, low-specificity splitter into a precision-controlled homogeneity test, at a modest cost in structured recall (0.92 to 0.81).

4.5 Geostatistical reference and the effective-sample-size approximation

Section 3.5 offers, alongside the default $1/e$-range proxy, a fitted variogram and its integral range, and Section 3.7 offers the exact Gaussian-simulation reference (11) in place of the effective-sample-size shortcut (6). Table 3 compares the three on one realisation of each world (six seeds, medians).

As a single homogeneity test the Gaussian-simulation reference (11) is exactly right: it accepts homogeneity on the class-free worlds ($p = 0.35, 0.46$) and rejects it on the structured and mixed worlds ($p < 10^{-3}$), with no effective-sample-size approximation. The constant κ also matters: the mean-based integral range (8) is some three times larger than ℓ^2 on these fields, so it is markedly

Table 3: The decorrelation-area constant and the exact null. $\hat{K}$ under the ℓ^2 proxy and under the variogram integral range; and the Gaussian-simulation homogeneity test (11) (p-value of the leading split, $B = 120$).

		$\hat{K}$		SGS null (11)
World	true K	$a = \ell^2$	integral range	p (decision)
Null	1	1	1	0.35 (K=1)
Trend	1	1	1	0.46 (K=1)
Struct	4	4	2	0.00 ($K \geq 2$)
Mixed	3	3	1	0.00 ($K \geq 2$)

more conservative, preserving specificity but losing recall (structured $\hat{K} = 2$, mixed $\hat{K} = 1$). The ℓ^2 proxy is, for the dip, the better calibrated of the two scalar choices. This is consistent with the caveat of Section 3.7: an integral range derived for the sample mean over-corrects a statistic of the empirical distribution, whose appropriate scale is the smaller indicator integral range.

Because the simulation reference (11) controls the level exactly for a single test on a full rectangular domain, one may hope to obtain a fully assumption-light estimator of K by using it at every node: that is, replacing the effective-sample-size quantile in Algorithm 1 by a fresh simulated quantile, computed from B Gaussian realisations fitted to each candidate cluster's own support. A direct implementation (SGS-Dip, $B = 40$) over-splits: specificity 0.85, structured accuracy 0.49, balanced score 0.535 against ESS-Dip's 0.771 (Table 2). This has two causes. First, the per-node test is repeated across the recursion without correction, so the per-node level no longer controls the family-wise false-split rate. Second, away from the leading split a node is a spatially scattered subset whose variogram must be fitted on a masked, non-rectangular support, biasing the simulated null.

The first cause is standard and correctable. A Bonferroni per-node level $\alpha/K_{\max}$, with the matching number of simulations, restores specificity to 0.99 and lifts the balanced score to 0.710, close to ESS-Dip's 0.771, the residual shortfall concentrated in the mixed regime (0.35 versus 0.50; SGS-Dip + correction, Table 2). So the simulation null is a viable recursive estimator once multiple testing is controlled. One aspect of the corrected results deserves a remark, because it can read as a paradox: the correction raises the exact-K accuracy on structured worlds (from 0.49 to 0.79), which seems to say that testing more conservatively finds *more* structure. It does not. The correction is uniformly conservative in the estimate itself, lowering the mean $\hat{K}$ in every world (from 5.02 to 4.33 on the four-class structured world, from 1.78 to 1.04 on the null); what rises is accuracy, because the uncorrected recursion overshoots the truth and pruning splits moves $\hat{K}$ toward K. Conservatism is monotone in $\hat{K}$; exact-K accuracy is not monotone in conservatism when the starting point over-detects. The second cause is more stubborn: on real imagery, where supports are irregular and non-stationary, even the corrected estimator trails ESS-Dip: single-class-window specificity 0.82 and 0.98 against 0.99 and 1.00 (Section 5.3), and over-detection of full scenes. The conclusion is therefore measured rather than categorical: the recursive simulation null becomes competitive on controlled fields once corrected, but its plug-in variogram on irregular supports, together with its cost of B simulations per node against the cached, near-linear effective-sample-size calibration, leaves (6) the better practical choice. We retain it as the method and use (11) as an assumption-light check on the leading split.

4.6 Computational performance

Figure 4 reports single-threaded wall-clock time on structured scenes. Across image sizes from $\sim$ 2,300 to $\sim$ 66,000 pixels, ESS-Dip scales with an empirical exponent $\mathrm{d}\log t/\mathrm{d}\log N = 0.86$ (panel a), near-linear, consistent with Section 3.10, and is a constant factor (about 1.2–2×) faster than the gap statistic, which scales similarly but carries the overhead of its reference replicates. Cost grows sub-linearly with the band count p (panel b: 0.31 s at $p = 3$ to 1.02 s at $p = 40$, at $N = 128^2$). Peak memory at $256 \times 256 \times 6$ is 8.9 MB, about three times the input cube, so the method is memory-light at scene scale. The exact Gaussian-simulation homogeneity test (11) with $B = 120$ costs 0.10–0.34 s over the same sizes, comparable to a single ESS-Dip evaluation, as it concerns one node; a fully recursive simulation null would multiply this by the number of nodes, which is the cost the effective-sample-size calibration is designed to avoid.

4.7 Robustness to anisotropy

The decorrelation area (7) is estimated from a radially averaged (isotropic) variogram, yet directional autocorrelation does not degrade the method. On null and structured worlds whose within-class range differs by a factor of up to eight between the two axes, ESS-Dip retains specificity 1.00 and structured accuracy 1.00 at every anisotropy ratio tested $(1, 2, 4, 8)$, whereas the gap statistic's specificity on the same class-free fields falls from 0.88 to 0.12 as the field elongates. The isotropic scalar is thus an adequate summary even under marked anisotropy, the modality signal it feeds being direction-agnostic, and a fully anisotropic variogram (Section 6) is a refinement rather than a necessity.

4.8 Robustness to misestimation of the range

The calibration consumes a single estimated scale, so the practical question is how much error in that scale the test tolerates. We answer it directly: the locally estimated range ℓ is multiplied by a factor $f \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \frac{3}{2}, 2, 4\}$ (the decorrelation area scales as f^2) and the full pipeline is re-run with the perturbed value, on the null world (specificity) and on the structured world with $K = 4$ (recovery), 15 paired scenes each. Table 4 reports the outcome. The level is essentially indifferent to the error: overestimating the range only makes the test more conservative, and even a four-fold *under*estimate, which inflates the effective sample size sixteen-fold, raises the false-split rate merely to 0.07, near the nominal α, because the least-favourable uniform calibration absorbs most of the slack (Remark 1). Recovery is asymmetric in the opposite direction: it survives underestimation unharmed (full recovery at $f = \frac{1}{4}$) and degrades only under overestimation, retaining $\hat{K} = 4$ in 73% of scenes at $f = \frac{3}{2}$ and collapsing to no splits at $f = 4$, where the effective sample size falls below the testable minimum. The deployed estimator therefore sits on the benign side of both asymmetries: errors of up to a factor of two in the range move the operating point along the precision–recall curve rather than breaking the calibration, and the failure mode of a gross overestimate is silence, not invention.

4.9 Class imbalance

The Sentinel-2 result (Section 5.4) suggests that a dominant class hides minorities; here we quantify how severe the imbalance must be. Two-class scenes are generated with a contiguous minority class occupying a controlled proportion $\pi \in \{0.50, 0.30, 0.20, 0.10, 0.05, 0.02\}$ of the domain (a Gaussian random field thresholded at its $(1 - \pi)$ quantile, class means drawn as in Section 4.1), 15 replicates at each of two separations (sep $= 3, 4$), pooled, with the same scene shown to every method.

Table 4: Sensitivity of the deployed pipeline to a multiplicative error f in the estimated range (15 paired scenes per world; area scales as f^2). $f < 1$ underestimates the range (inflates n_{eff}); $f > 1$ overestimates it.

Range factor f	1/4	1/2	3/4	1	3/2	2	4
Null world: false-split rate	0.07	0.00	0.00	0.00	0.00	0.00	0.00
Structured (K=4): mean $\hat{K}$	4.00	3.93	3.93	3.80	3.67	2.80	1.00
Structured (K=4): $\mathbb{P}(\hat{K}=4)$	1.00	0.93	0.93	0.80	0.73	0.47	0.00

Table 5: Detection of a contiguous minority class of proportion π (true $K = 2$; 30 pooled scenes per cell, identical scenes across methods). For the baselines, $\mathbb{P}(\hat{K} \geq 2)$ overstates detection because their false-split rate on class-free scenes is near one (Table 2).

	ESS-Dip		ESS-Dip-R		Gap statistic		Silhouette	
π	$\mathbb{P}(\hat{K}\geq 2)$	$\bar{\hat{K}}$	$\mathbb{P}(\hat{K}\geq 2)$	$\bar{\hat{K}}$	$\mathbb{P}(\hat{K}\geq 2)$	$\bar{\hat{K}}$	$\mathbb{P}(\hat{K}\geq 2)$	$\bar{\hat{K}}$
0.50	0.97	1.97	0.97	1.97	1.00	2.00	1.00	2.00
0.30	1.00	2.00	1.00	2.00	1.00	2.00	1.00	2.00
0.20	0.90	1.90	1.00	2.00	1.00	2.10	1.00	2.00
0.10	0.73	1.73	0.93	1.93	0.97	2.07	1.00	2.00
0.05	0.00	1.00	0.50	1.50	0.87	2.33	1.00	2.03
0.02	0.00	1.00	0.00	1.00	0.40	1.60	1.00	2.97

Table 5 reports the minority-detection rate $\mathbb{P}(\hat{K} \geq 2)$ and the mean $\hat{K}$. ESS-Dip detects the minority reliably down to $\pi = 0.20$ (0.90), degrades at $\pi = 0.10$ (0.73) and is blind below $\pi = 0.05$; the recall variant ESS-Dip-R extends the reliable regime to $\pi = 0.10$ (0.93) and detects half the cases at $\pi = 0.05$. The mechanism is the calibration itself: a minority occupying proportion π contributes a mode of mass π to the projected distribution, and once $\pi\, n_{\text{eff}}$ falls to the order of the decorrelation area the excess mass is indistinguishable from autocorrelated fluctuation. The apparent robustness of the baselines at extreme imbalance is not creditable as detection: their false-split rate on class-free scenes is near one (Table 2), so a reported $\hat{K} \geq 2$ carries no evidence, and at $\pi = 0.02$ the silhouette returns $\hat{K} \approx 3$ by fragmenting the majority class rather than isolating the minority. As practical guidance: minorities below about a fifth of the scene call for ESS-Dip-R, and minorities below a twentieth are beyond the specificity-controlled family at this separation, a limitation Section 6 returns to.

5 Application to land-cover imagery

We now turn to real scenes, where the question is no longer whether a method recovers a known K but whether it behaves sensibly on data with genuine spatial autocorrelation, smooth illumination gradients, and spectrally overlapping classes. We use three benchmarks spanning two sensing modalities and report two experiments: a direct test of the specificity property of Section 3.9 on regions known to contain a single class, and a full-scene estimate of the number of classes.

5.1 Data and preprocessing

Two airborne hyperspectral scenes with per-pixel ground truth are used: the *Indian Pines* (145×145, 200 bands after removal of water-absorption bands, 16 classes; Baumgardner et al., 2015) and the *Salinas* scene (512×217, 204 bands, 16 classes), both standard in the classification literature. Each cube is standardised per band and reduced by principal components to ten dimensions, retaining 95.5% and 99.6% of the variance respectively. As a multispectral counterpart, the common satellite setting, we use a 246×237 Sentinel-2 Level-2A scene over the Po Valley with ten surface-reflectance bands (B02–B12), labelled by the ESA WorldCover product (Zanaga et al., 2022); four classes exceed 1% of the scene (crop 83%, tree 9%, grass 5%, built 4%), so its nominal number of classes is four. Bands are standardised; no dimensionality reduction is needed.

5.2 Within-class autocorrelation

The calibration presumes a within-class autocorrelation that is real, estimable, and separable from the between-class mosaic; we verify all three on the data. Averaging the radial empirical semi-variogram of the first principal component over pure single-class windows, each scene shows a well-defined within-class structure (Figure 5): an exponential model fits Indian Pines and Salinas (ranges 1.8 and 1.4 pixels) and a spherical model fits Sentinel-2 (range 5.4 pixels), with $1/e$ correlation lengths of two to three pixels. The indicator-to-field integral-range ratio derived in Section 3.7 holds on these real variograms: $a_{\text{int}}/a_I = 1.51$, 1.54 and 1.51 for the three scenes, close to the 1.45 of the Gaussian fields. Local estimation separates the within-class range from the scene scale: the tile-median range is 3 to 4 pixels, whereas the scene-wide estimate, conflating the between-class mosaic, is three to fifteen times larger (14.5 to 46 pixels). The premise that most tiles fall within a single class holds on the contiguous scenes (62% of Salinas and Sentinel-2 tiles are single-class) and weakens on the fragmented Indian Pines fields (27%), where the tile median nonetheless recovers a within-class scale (3.5 against the 16-pixel scene range), the robustness for which the median is chosen.

5.3 Specificity on single-class regions

A region covered by a single land-cover class is the real-data analogue of the null and trend worlds of Section 4: its pixels are autocorrelated and spectrally variable (soil, moisture and illumination vary within a crop field), yet the true number of classes is one. We extract all 16×16 windows in which a single ground-truth class covers at least 90% of the pixels (67 windows across Indian Pines and Salinas, 60 in the Sentinel-2 scene, all of the dominant crop class), and record the number of clusters each method reports. Table 6 summarises the outcome; the proposed-family ablations of Section 4.2 all behave like ESS-Dip here (specificity ≥ 0.97) and are omitted for brevity.

The simulation result reproduces on real imagery. The silhouette, Calinski–Harabasz and Davies–Bouldin indices split a single real land-cover class into two or more sub-clusters in *every* window of both modalities; the gap statistic does so in all but the cleanest hyperspectral windows ($\hat{K}$=1 rate 0.64) and, on the multispectral crop fields, fragments a single class into nearly seven on average. ESS-Dip identifies the region as a single class in 99% of the hyperspectral and 100% of the multispectral windows. The within-field spectral variability that the standard indices read as structure is precisely what the effective-sample-size calibration discounts. The two closer baselines confirm the attribution on real data: the Hennig–Lin spatial null, calibrated for autocorrelation, improves on the standard indices over the hyperspectral windows (0.72) but collapses on the crop-dominated Sentinel-2 fields (0.07, mean $\hat{K} = 4.8$), where within-field variation defeats its phase-randomized reference; plain dip-means, the modality test without the effective-sample-size calibration, reaches 0.84 and 0.63. Both trail ESS-Dip's 0.99 and 1.00, isolating the contribution of the spatial calibration on real imagery. The result is not an artefact of the preprocessing: on the Indian Pines windows ESS-Dip returns a single class in every window across principal-component

Table 6: Number of clusters reported on single-class windows (true $K = 1$). A method calibrated for spatial autocorrelation should return $\hat{K} = 1$.

	Hyperspectral (67 win.)		Sentinel-2 (60 win.)	
Method	$\hat{K}$=1 rate	mean $\hat{K}$	$\hat{K}$=1 rate	mean $\hat{K}$
Gap statistic	0.64	1.79	0.00	6.97
Silhouette	0.00	2.27	0.00	2.35
Calinski–Harabasz	0.00	2.70	0.00	3.20
Davies–Bouldin	0.00	3.37	0.00	2.62
Hennig–Lin (spatial)	0.72	2.36	0.07	4.77
Dip-means (nominal)	0.84	1.24	0.63	1.55
ESS-Dip	**0.99**	**1.01**	**1.00**	**1.00**
ESS-Dip-R (recall)	0.99	1.01	1.00	1.00

truncations from five to thirty components, window sides from 12 to 24 pixels, and purity thresholds from 0.85 to 0.95 (for those combinations that yield windows). The recursive Gaussian-simulation estimator, even with the multiple-testing correction of Section 4.5, is less reliable here: it returns a single class in 0.82 of the hyperspectral and 0.98 of the multispectral windows, below ESS-Dip, because its per-node plug-in variogram degrades on real, non-stationary support, the residual effect that the correction does not remove.

5.4 Full-scene estimation

Applied to an entire labelled scene, the methods estimate the number of land-cover classes present (Table 7). Two caveats frame the reading of the table. First, $\hat{K}$ estimates the number of classes *distinguishable from the autocorrelation null at the effective sample size*, not the nominal label count; failing to reject does not certify that finer labelled classes are absent, only that the data do not support them against the null, so a testing-based $\hat{K}$ is conservative by construction. Second, no criterion approaches the nominal sixteen classes of the hyperspectral scenes: the labels include many small, spectrally similar agricultural classes that are not separable without supervision, and the best standard index reaches nine (the gap on Salinas) while the others return two to six. ESS-Dip is deliberately conservative, reporting two classes on Indian Pines and five on Salinas. On the crop-dominated Sentinel-2 scene, where a single class covers 83% of the pixels, ESS-Dip returns that one dominant class, whereas the gap reports five. The recall variant ESS-Dip-R recovers more where the structure is separable, reporting eleven classes on Salinas at the same single-class-window specificity (Table 6), while remaining conservative where it cannot help: two on the spectrally overlapping Indian Pines and one on the imbalanced Sentinel-2, which are limited by separability and dominance, not by the calibration's conservatism.

These full-scene figures should be read against the precision–recall trade-off of Section 4.3. Exhaustive recovery of many overlapping classes from a single scene is beyond every criterion considered, and the conservatism of ESS-Dip is the price of its specificity: a procedure that refuses to invent structure will, on a scene dominated by one class or composed of weakly separated ones, under-report rather than over-report. The recursive simulation estimator errs in the opposite direction, over-detecting to the imposed cap on Salinas and to six classes on the crop-dominated Sentinel-2 scene even after the multiple-testing correction, the plug-in variogram on the scene's irregular, non-stationary support compounding over a full-scene recursion. We regard the method's

Table 7: Estimated number of classes on full scenes; the nominal number of labelled classes is given in parentheses. The ≥ 20 entry marks the recursive simulation estimator reaching its recursion cap without terminating.

Method	Indian Pines (16)	Salinas (16)	Sentinel-2 (4)
Gap statistic	3	9	5
Silhouette	2	2	2
Calinski–Harabasz	2	6	3
Davies–Bouldin	2	6	2
ESS-Dip	2	5	1
ESS-Dip-R (recall)	2	11	1
SGS-Dip (corrected)	2	≥ 20	6

Table 8: Partition quality on the full scenes: adjusted Rand index (ARI) and normalised mutual information (NMI) against the ground-truth land-cover map, with the number of classes each method returns. Computed on labelled pixels.

Scene	Method	$\hat{K}$	ARI	NMI
Indian Pines	Gap statistic	3	0.21	0.36
	Silhouette	2	0.18	0.37
	ESS-Dip	2	0.18	0.37
Salinas	Gap statistic	9	**0.52**	**0.71**
	Silhouette	2	0.18	0.37
	ESS-Dip	5	0.32	0.57
Sentinel-2	Gap statistic	5	0.10	0.14
	Silhouette	2	0.36	0.22
	ESS-Dip	1	0.00	0.00

value as residing in that precision (it answers reliably whether genuine class structure is present beyond spatial autocorrelation) rather than in maximal class enumeration, and we return to this distinction, and to ways of relaxing it, in Section 6.

The number of classes is only half the picture; the quality of the induced partition completes it. Table 8 reports the adjusted Rand index (ARI) and normalised mutual information (NMI) of each labelling against the ground-truth map. Two things stand out. First, unsupervised recovery of these partitions is hard for every method: no ARI exceeds 0.52. Second, on the multi-class hyperspectral scenes ESS-Dip's partition quality trails the over-detecting gap statistic (Salinas ARI 0.32 versus 0.52), and on the crop-dominated Sentinel-2 scene, where it returns a single class, its ARI is zero. This is the partition-quality counterpart of the conservatism already visible in the class counts: a precision-oriented test that refuses to split does not, and is not meant to, maximise partition recovery. The numbers make the trade-off explicit and reinforce the reading of ESS-Dip as a homogeneity guard rather than a classifier.

6 Discussion

A precision-oriented criterion. The results of Sections 4 and 5 draw a consistent picture across synthetic fields, hyperspectral scenes and multispectral imagery: standard internal criteria are powerful at recovering well-separated classes but cannot recognise their absence, whereas ESS-Dip controls the false-split rate at the cost of recall when classes are many or weakly separated. The two behaviours are not interchangeable failures of calibration but the two ends of a deliberate trade-off. By referring the dip statistic to the effective rather than the nominal sample size, the test is made conservative in exact proportion to the information the autocorrelation removes; the level α and the range ℓ are the knobs that set where on the precision–recall curve the method sits. We have fixed them once, without per-scene tuning, and the resulting operating point is one of high precision: ESS-Dip answers reliably whether discrete structure is present *beyond* spatial autocorrelation, and is correspondingly cautious about enumerating structure that the data do not independently support. For the analyst this is the useful direction of error, a guard against the over-segmentation that Section 5.3 documents for the standard indices, but it is a genuine limitation when an exhaustive class inventory is the goal. This fixes the method's natural role: not a stand-alone classifier (on real multi-class scenes it under-reports), but a calibrated homogeneity test, a guard placed before an unsupervised classification or a stopping rule for a region-growing or regionalization procedure (Section 2.4), where a negative one can trust is worth more than an exhaustive but unreliable class count.

Choosing within the family, and embedding it in practice. The two calibrations serve different users. The default ESS-Dip is the choice when a false split is the costly error (a homogeneity guard before classification, a stopping rule, a confirmatory analysis) and whenever the within-class marginal may be non-Gaussian, since the uniform null is distribution-free. ESS-Dip-R is the better general-purpose class enumerator when approximate within-class Gaussianity is plausible, as it typically is after per-band standardisation and principal-component reduction: it recovers substantially more structure at the same measured specificity (eleven against five classes on Salinas; balanced score 0.868 against 0.771). The imbalance sweep of Section 4.9 makes the boundary quantitative: minority classes below about a fifth of the scene call for the recall variant, and below a twentieth lie beyond both. Neither variant needs a pre-selected univariate feature: the projection is part of the procedure, computed in the full feature space, and what the method does require is a space in which Euclidean geometry is meaningful, so bands should be standardised and strongly collinear cubes reduced by principal components (we use ten throughout; Section 5.3 shows the results are insensitive to truncations from five to thirty). Three embedding patterns cover the practical uses: as a pre-test that decides whether any unsupervised classification of a scene is warranted at all; as the stopping rule of a divisive or region-growing procedure, including spatially constrained clustering (Section 2.4); and as a post-hoc audit that applies the homogeneity test within each cluster proposed by any external algorithm, flagging the clusters the data do not support.

Relation to geostatistical null models. Assumption 1 replaces the independence reference of the classical criteria by a unimodal Gaussian random field, the continuous-field counterpart of the spatial neutral models used in disease mapping (Goovaerts and Jacquez, 2004) and of Moran spectral randomization (Wagner and Dray, 2015). We realise it implicitly, through the effective-sample-size calibration of the dip, rather than by simulating reference fields. A more explicit route would calibrate the statistic against an ensemble of sequential-Gaussian-simulation realisations conditioned on the empirical variogram; this would relax the isotropy built into our scalar range

ℓ and accommodate anisotropic or nested autocorrelation structures at the cost of the simulation effort. We see the present scalar calibration and a full simulation-based null as two points on a spectrum trading fidelity against computation, and the near-linear cost of the former (Section 3.10) as its principal practical advantage on megapixel scenes. The choice of null also encodes an analytical judgement rather than a universal standard: the family's reference declares autocorrelated continuous variation to be non-structure, which is the right convention for class counting, but applications in which smooth sub-structure is itself of interest (gradients within a single cover type that an analyst wishes to subdivide, say) should not demand that their clusters of interest defeat an autocorrelation null.

Limitations. Four limitations bound the method's applicability. First, when many spectrally overlapping classes coexist, as in the sixteen-class hyperspectral scenes, no unsupervised criterion, ours included, approaches the nominal class count (Section 5.4); recovering such inventories requires supervision or auxiliary information that an internal criterion cannot supply. Second, strong class imbalance degrades recall: when one class dominates the scene, as the crop class does in the Sentinel-2 image, the leading split direction is governed by that majority and minority classes are not resolved. Third, the false-split control of Proposition 1 is an asymptotic statement resting on the empirical-process hypothesis (C3) and a fixed projection, not a finite-sample guarantee. Fourth, the 2-means split direction is matched to balanced, spherical groups rather than to density modes: when modes carry very different variances, or more than two classes coexist at a node, the proposed direction can fail to expose the multimodality or can cut through a class. The recursion re-tests every child, and a poor direction costs recall rather than level (Section 3.3); mode-seeking or projection-pursuit directions are the natural refinement of this heuristic.

Extensions. Each limitation suggests a direction. The recall-oriented variant ESS-Dip-R realises one: trading the distribution-free uniform null for the model's Gaussian one recovers more structure at the same specificity (eleven classes on Salinas against five). Further recall could come from revisiting leaves at a relaxed level, or testing minority modes against the residual of the majority, to mitigate the imbalance that bounds the Sentinel-2 result. Replacing the scalar range by an anisotropic variogram model, and the uniform dip null by conditional simulation, would extend the calibration to non-stationary autocorrelation, although Section 4.7 shows the isotropic estimate is already robust to directional autocorrelation up to an eight-fold anisotropy ratio, in the scenarios simulated there. The same effective-sample-size correction applies, unchanged, to spatio-temporal stacks, where the decorrelation "area" becomes a space–time volume. Finally, because our contribution is a stopping rule rather than a partitioning constraint, it composes naturally with spatially constrained clustering (Section 2.4): the calibrated test could decide when a regionalization has produced as many regions as the data independently support.

7 Conclusions

Selecting the number of clusters on spatial data is compromised by a property that the standard criteria share and that spatial data expose: they take no account of the dependence between observations. We have shown that the omission is not benign. Under spatial autocorrelation the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices over-detect without exception, inventing classes on featureless fields and fragmenting single land-cover regions on real imagery.

We have proposed a remedy framed in geostatistical terms. Recast as a calibrated test of cluster homogeneity, asking whether a region carries multimodality in excess of spatial autocorrelation, ESS-Dip evaluates the Hartigan dip along the locally optimal split direction and calibrates it against the *effective* sample size, estimated from a within-class autocorrelation range obtained by local range estimation, after removing a low-order trend whenever one is resolvable. The procedure carries an asymptotic false-split-control property and runs in near-linear time. On a factorial simulation with known ground truth it reaches unit specificity, and its recall variant ESS-Dip-R attains the best balanced score of sixteen criteria at the same specificity; on the Indian Pines, Salinas and Sentinel-2 scenes both identify single land-cover regions as homogeneous where the standard criteria do not. Their reliable negatives suit the family to use as a homogeneity guard or a stopping rule for region-growing as much as to cluster-number selection, spanning a precision–recall axis from the conservative distribution-free null to the higher-recall Gaussian one.

The broader message is that number-of-clusters selection on autocorrelated data must be calibrated to the information the data actually contain, not to their nominal size, and that the signal distinguishing classes from autocorrelation is modality, not dispersion. The effective sample size and declustering, long-standing tools of geostatistics, supply exactly what is needed to make a classical clustering criterion honest about spatial dependence. All code and data required to reproduce the simulation study and the applications are released.

Code and data availability

All code required to reproduce the simulation study and the applications (the synthetic-scene generators, the implementation of ESS-Dip and of every baseline and ablation, the benchmark driver, and the figure scripts) is openly available at `https://github.com/franciscoparrao/ess-dip`, released under the MIT licence, and will be archived on Zenodo with a permanent DOI on acceptance. The Indian Pines and Salinas hyperspectral scenes are the standard public benchmarks (Baumgardner et al., 2015); the Sentinel-2 Level-2A imagery is produced by the Copernicus programme and was accessed through the Microsoft Planetary Computer; the land-cover reference is the ESA WorldCover product (Zanaga et al., 2022). The synthetic scenes are generated deterministically by the released code from the seeds reported in the repository.

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Declarations

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Conflict of interest. The author is the sole author of this work and declares that he has no competing interests relevant to its content.

Data and code availability. All code and data required to reproduce the study are openly available, as detailed in the section "Code and data availability".

A Implementation specification

This appendix specifies the deployed procedure completely, so that it can be re-implemented from the paper alone; the released code (Section "Code and data availability") is the executable reference and fixes every random seed used in the experiments.

Inputs and preprocessing. The method takes a feature image $\boldsymbol{z}$ on a lattice D (any p); no preprocessing is part of the method itself. In all real-data experiments the bands are standardised to zero mean and unit variance per band, and the hyperspectral cubes are reduced by principal components to $p = 10$ (Section 5.1); Section 5.3 reports the insensitivity of the results to that truncation (5 to 30 components).

Fixed constants. All constants are fixed once, across every experiment in the paper; none is tuned per scene.

Symbol	Value	Role	Where it enters
α	0.05	test level	Eq. (6)
τ	0.25	detrend threshold	Eq. (10)
–	2	polynomial degree of the drift	Eq. (9)
t	24	tile side in pixels (reduced to $\min(24, \lfloor H/2 \rfloor, \lfloor W/2 \rfloor)$ on small domains)	Eq. (7)
ν	12	minimum effective size to test a node	Algorithm 1
B	5	ensemble size (median of $\hat{K}$)	Section 3.8
–	300	Monte Carlo draws per null quantile	Section 3.4
–	$[12, 2000]$	clamp of the rounded n_{eff} before quantile lookup	Section 3.4
–	8 / 20	recursion cap on $\hat{K}$ (factorial study / full scenes)	Algorithm 1

Null quantile. For a node with effective size n_{eff}, let $m = \text{clamp}(\lfloor n_{\text{eff}} \rceil, 12, 2000)$. The critical value $q_{1-\alpha}(m)$ is the empirical $(1-\alpha)$ quantile of 300 dip statistics, each computed from m pseudo-random draws of the reference null: $\text{U}(0,1)$ for ESS-Dip (least-favourable, distribution-free) and $\text{N}(0,1)$ for ESS-Dip-R (the null of Assumption 1). Quantiles are memoised by m (and by reference law), so each distinct size is simulated once per run.

Split proposal. At a node C, 2-means is run on the feature vectors of C (k-means with $k = 2$, five restarts); the split direction $\boldsymbol{u}$ is the normalised difference of the two cluster means, Eq. (4), and the dip is computed on $\{\boldsymbol{u}^\top \boldsymbol{z}(s)\}_{s \in C}$ by the algorithm of Hartigan and Hartigan (1985). A significant test splits C into the two 2-means children, each retained only if its pixel count is at least the decorrelation area a (so that $n_{\text{eff}} \geq 1$).

Range estimation. The decorrelation area $a = \ell^2$ is estimated once per scene (not per node), on the detrended field when Eq. (10) activates: per tile, the first principal component's sample autocorrelation is computed by FFT and its $1/e$ crossing recorded, averaged over the two axes; ℓ is the median over tiles, Eq. (7).

Ensemble. The recursion is run $B = 5$ times with independent seeds (the only stochastic elements are the k-means initialisations and the null-quantile Monte Carlo); the reported $\hat{K}$ is the ensemble median.

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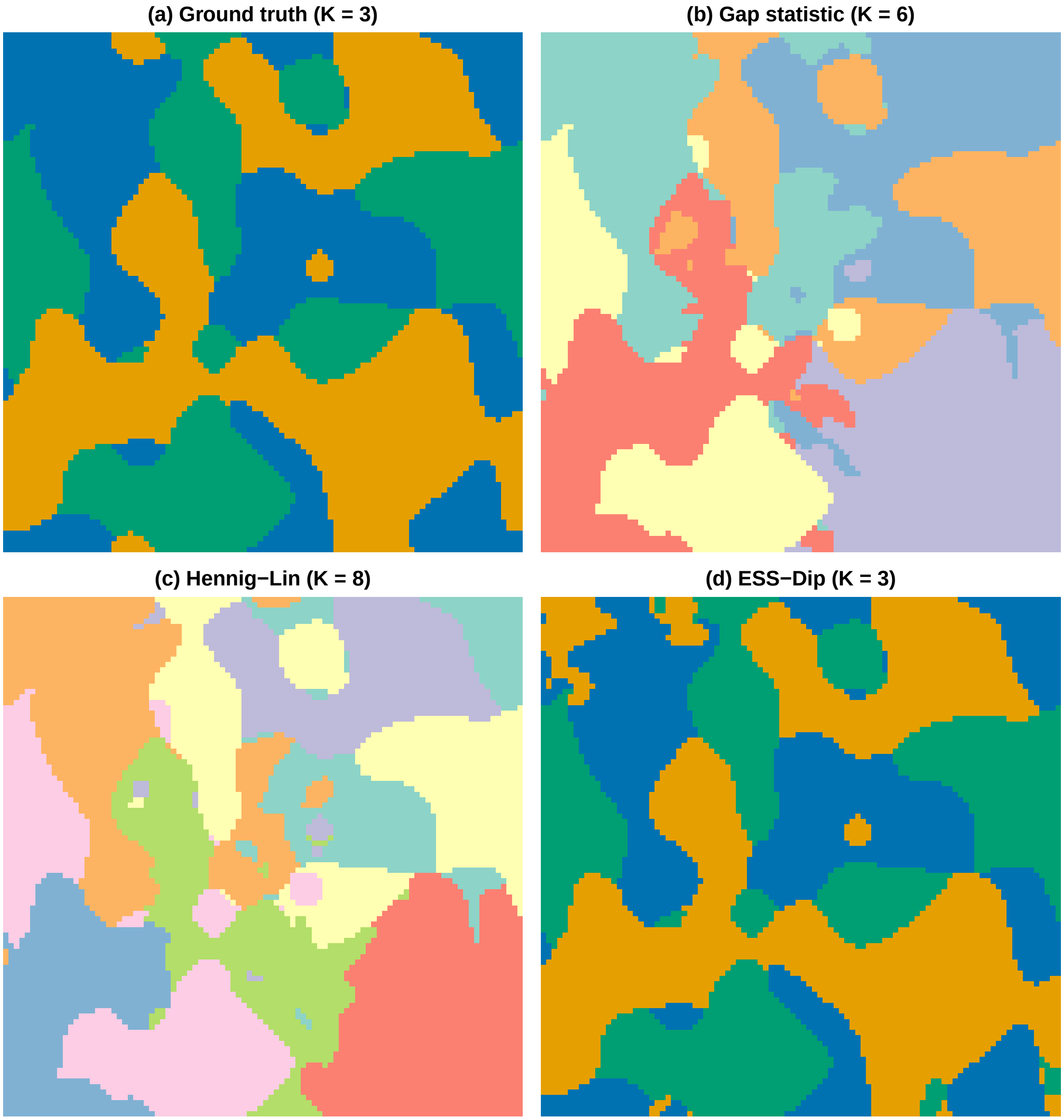

Figure 3: Recovery on a representative mixed scene (three classes plus a smooth trend). The gap statistic over-segments into six clusters (b) and the Hennig–Lin spatial null into eight (c), both reading the trend and the within-class autocorrelation as extra structure; ESS-Dip removes the trend and calibrates to the effective sample size, recovering the three true classes (d), whose partition matches the ground truth (a). Cluster colours in (b) and (c) are arbitrary; in (a) and (d) they are matched, so a correct recovery reproduces the colours of panel (a).

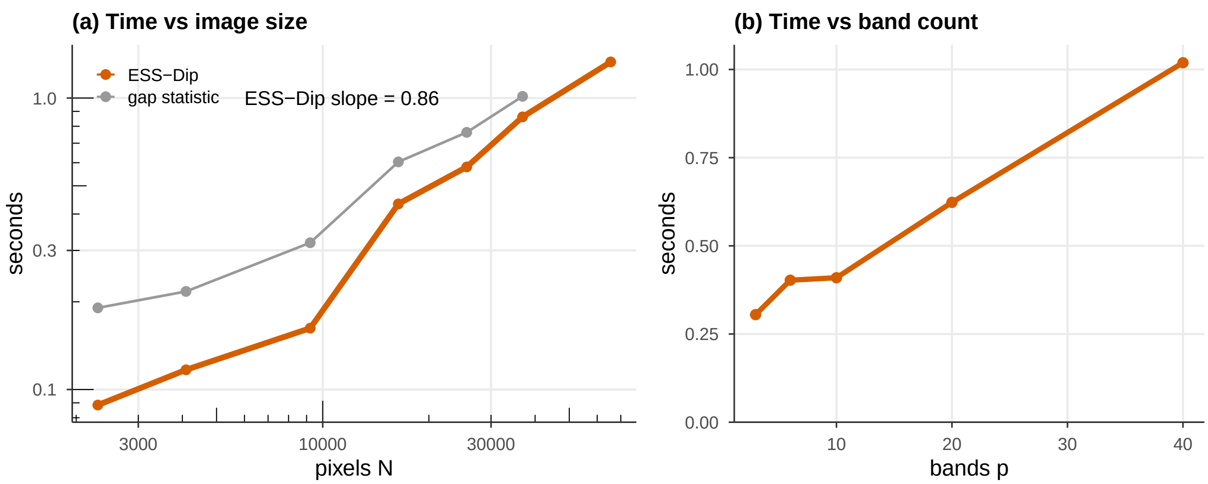

Figure 4: Computational performance (single-threaded). (a) Wall-clock time versus pixel count N on a log–log scale, with the fitted ESS-Dip scaling exponent; (b) time versus band count p at $N = 128^2$.

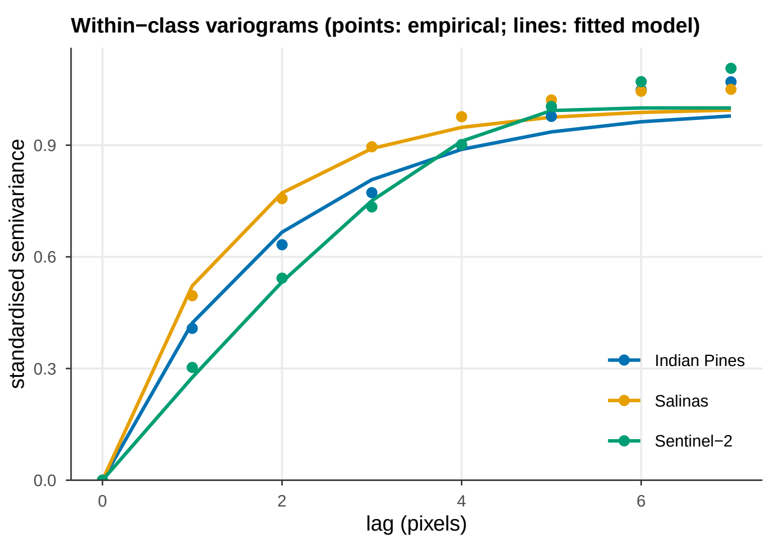

Figure 5: Within-class variograms on the real scenes: the radial empirical semivariogram of the first principal component (points), averaged over pure single-class windows and standardised to unit sill, with the best-fitting exponential, spherical or Gaussian model (lines). The within-class autocorrelation the calibration relies on is real and has a well-defined range on actual imagery.